

YU-HSUAN HUANG

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PROFESSIONAL EXPERIENCE

Shin Fang Co., Ltd.

Robotics Engineer

Taipei, Taiwan

Oct 2023 – Aug 2026

- Established an end-to-end imitation learning pipeline, including data collection, training, and model deployment in both simulation and real-world environments, resulting in successful autonomous behavior in practical tests.
- Engineered a low-latency teleoperation system, working on both hardware and software architecture, enhancing control algorithms, and testing multiple strategies, which enabled smooth and accurate data collection.
- Built a planning and control framework for a dual-manipulator kitchen robot, enabling flexible switching between various end-effectors. Successfully deployed in both Isaac Sim and real hardware setups.
- Created a vision-based bin-picking function, using a foundation model backend and improving picking consistency in cluttered, unstructured environments.
- Integrated a custom electronic stove control unit for automated flame control, improving system responsiveness and task stability under real-world kitchen conditions; named inventor on US patent application 2026/0110436 A1.
- Enabled commercialization of the company's dual-arm robot by delivering a low-cost hardware version, at about one-fifth the original hardware cost, that ran the same planning and learning software stack.

Synopsys Taiwan Co., Ltd.

R&D Engineer

Taipei, Taiwan

Jul 2022 – Jul 2023

- Improved validation pipelines for IC Validator, Synopsys's physical verification tool, boosting testing quality and meeting industry-standard benchmarks, helping the team earn the Software Quality Award for reliable, on-time delivery.

Brogent Technology, Inc.

Advisory Software Engineer

Kaohsiung, Taiwan

Jan 2022 – Jun 2022

- Independently designed and built the full hardware and software stack for an autonomous mobile robot, delivering a working prototype that demonstrated technical feasibility to executives.

International Center of Excellence in Intelligent Robotics and Automation Research

Research Specialist

Taipei, Taiwan

Dec 2018 – Oct 2021

- Designed gravity compensation algorithms to enable intuitive teach-and-play control on an industrial robot arm, improving ease of use and increasing operator safety in manual programming scenarios.
- Formulated a deterministic real-time trajectory generation algorithm running at a 1 kHz control loop, ensuring smooth, responsive robot movement.
- Integrated a robot arm with an AOI machine, deploying a pick-and-place cell with optical sensors to automate spray-gun quality checks and reduce human inspection errors.
- Implemented the control system and navigation algorithm for an 8-DOF swerve-drive mobile robot, enabling agile, omnidirectional movement with high precision in dynamic environments.
- Built a 3D-printed six-axis robot arm from scratch with 1 kg payload, assembling mechanical components, followed by developing a motion planning framework to enable essential robotic arm movements and control.

TECHNICAL SKILLS

ROS / ROS2 / Linux / C++ / Python / Docker / NVIDIA Isaac Sim / LeRobot / Git / SolidWorks / MATLAB

EDUCATION

University of Michigan – Ann Arbor

M.S. in Mechatronics, Robotics, and Automation Engineering

Ann Arbor, USA

Sep 2026 – Present

National Taiwan University

M.S. in Mechanical Engineering, Mechanism and Machine Theory Lab

Taipei, Taiwan

Sep 2015 – Jul 2017

National Taiwan University

B.S. in Civil Engineering

Taipei, Taiwan

Sep 2012 – Jul 2015